

Chris Zhang

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EDUCATION

Northeastern University, College of Engineering

Master of Science Degree in Mechanical Engineering, Mechatronics Concentration

Courses: Advanced Mechanics of Materials, Materials Process Selection, Math Methods, Mechatronic Systems, Control Systems, Scientific Machine Learning, Power Electronics, Wearable Robotics

Extracurriculars: ASME, Club Ultimate Frisbee Captain

Boston, MA

April 2026

GPA: 3.63

Bachelor of Science Degree in Mechanical Engineering

Awards: Dean's List (Fall 2021, Spring 2022, Spring 2024, Spring 2025), Dean's Scholarship

Courses: Heat Transfer, Systems and Controls, Computation, Design, Measurements, Thermodynamics, Mechanics of Materials, Statics, Intro to Material Science, Dynamics, Fluid Mechanics, Electrical Engineering, Fundamentals of Computer Science 1 and 2, Capstone

Extracurriculars: Society of Asian Scientists and Engineers, Electric Racing, ASME

May 2025

GPA: 3.5

EXPERIENCE

ICO Energy and Engineering, Boston, MA

July 2024 – December 2024

Commissioning Engineer

- Executed comprehensive commissioning tests on HVAC penthouses across 6+ commercial facilities, ensuring safety compliance and system operates according to specifications
- Collected pipe flow and energy data on site for meter allocation services, highlighting faulty meters that needed calibration to correct building energy use data to reduce excess energy spending by ~15%
- Modeled building energy usage intensity using energy and water billing to gauge potential ~30% savings from commissioning work for a potential client

Instron, Norwood, MA

July 2023 – December 2023

Applications Engineer

- Determined optimal tensile test equipment based on knowledge of ISO and ASTM standards for specific client applications regarding plastics, metals, thin films, elastomers, and more
- Generated reports sent to high profile clients detailing recommended testing frames, fixtures, and extensometers, assisting with orders up to \$40,000+
- Engaged in cross-functional collaboration to generate testing report template for 3D printed plastic samples to be employed for potential clients seeking 3D printed applications
- Employed digital image correlation on electric car battery thin films and mapped full-field strain and displacement images in Blue Hill Universal for applications team's future customer demonstrations

PROJECTS

Northeastern University, Boston, MA

January 2026 – April 2026

Canine Rehabilitation Wheelchair, Wearable Robotics

- Collaborated with group of 4 to create an assistive mobility device for hind-leg paralyzed canines that generated passive range of motion during wheelchair locomotion
- Designed and fabricated 3D-printed pedal assemblies in SolidWorks, iterating designs based on patient-specific measurements and constraints from crankshaft rotation
- Processed IMU trial data in MATLAB using high-pass filtering and kinematic analysis to quantify hip, knee, and back articulation, validating benefits of pedal-driven hind leg motion

Northeastern University, Boston, MA

June 2024 – April 2025

Auto Return Trainer, Mechanical Engineering Capstone

- Co-led team of 6 to design and fabricate an automated soccer ball launcher delivering game-realistic shots at speeds up to 70 mph for goalkeeper training
- Designed motor mounts and wheel axle adapters in SolidWorks with $\pm 0.005"$ tolerances and GD&T specifications, ensuring precision alignment for consistent ball launch trajectories
- Managed \$3000 project budget through Excel materials ordering sheet and coordinated testing facility reservations and measuring equipment availability with Capstone advisors

SKILLS

CAD & Design: SolidWorks (CSWA Certified), AutoCAD, Abaqus, GD&T, DFM

Computer & Electrical: Excel, PowerPoint, Arduino, Digital Multimeter, Oscilloscope, LTspice, PLECS

Fabrication & Prototyping: 3D Printing, Bench Assembly, Hand Tools, Soldering, Sensor Integration

Programming Languages: MATLAB, Python C++, Java, JavaScript, C#